

1 Topics Covered

- Space Complexity Basics
- Configurations and Space Bounds
- Relativization and Oracles
- NSPACE and the Space Hierarchy Theorem
- PSPACE

2 Space Complexity Fundamentals

We relate standard deterministic time to space and nondeterministic space as follows:

$$\text{DTIME}(S(n)) \subseteq \text{SPACE}(S(n)) \subseteq \text{NSPACE}(S(n)) \subseteq \text{DTIME}\left(2^{O(S(n))}\right)$$

Where $S : \mathbb{N} \rightarrow \mathbb{N}$ and the language $L \in \{0, 1\}^*$.

Definition: $L \in \text{SPACE}(S(n))$ if there exists a constant c and a Turing Machine M such that at most $c \cdot S(n)$ locations on M 's work tape are visited during its computation on every input of length n .

We generally assume $S(n) \geq \log(n)$. In order to visit $S(n)$ spaces, a machine requires at least $\text{DTIME}(S(n))$ time just to move the head.

Configurations

How can $\text{DTIME}(2^{O(S(n))})$ bound space? We look at the total number of unique machine configurations. A configuration C consists of:

- **Head:** Current state of the finite control and the positions of the head on both the input tape and work tapes.
- **Work Tape:** The exact contents written on the $O(S(n))$ cells of the work tape.

Because the alphabet and state space are finite, the size of a single configuration is $O(S(n))$. The total number of possible distinct configurations is bounded by $2^{O(S(n))}$.

for tree= grow'=0, minimum size=2ex, inner sep=4pt $[C_{start} [C_1^1, \text{edge label=node[midway, above] } \delta_0 \dots [C_{accept}]]] [C_2^1, \text{edge label=node[midway, above] } \delta_1]]$

Every edge in the configuration graph denotes a valid transition function. A nondeterministic machine may have multiple outgoing transitions (forks) from a single configuration. To recall the current status of computation, the machine must solely recall the current configuration.

3 Relativization and Oracles

Oracle Aided Classes: P^O, NP^O , where an oracle is a function $O : \{0, 1\}^* \rightarrow \{0, 1\}$. An algorithm can use its oracle to solve function O "magically".

Example: Let $O = \text{HALT}$. We then have:

$$P^{\text{HALT}} \supsetneq NP, EXP, NEXP$$

We know that $\text{DTIME}(f(n)) \subsetneq \text{DTIME}(g(n))$ where $f(n) \log(f(n)) = o(g(n))$. This separation holds for *any* oracle O . This means that although set of uncomputable languages is large, with strictly more time, you can solve strictly more problems.

3.1 Can we use diagonalization to prove $P \neq NP$?

No. Diagonalization only separates classes built on the *same* resource constraints.

Theorem: There exist oracles $A, B : \{0, 1\}^* \rightarrow \{0, 1\}$ such that $P^A = NP^A$, but $P^B \neq NP^B$.

Because standard diagonalization proofs relativize (meaning they hold regardless of the oracle used), any "Black Box" diagonalization technique cannot resolve $P \neq NP$. If a proof method works for all oracles, it will contradict either Oracle A or Oracle B.

4 Nondeterministic Space (NSPACE)

When measuring space:

- **Input tape** [] $\leftarrow$ Squares used here are **not** counted.
- **Work tape** [] $\leftarrow$ Only squares touched here are counted.

We do not bound the running time, only the space.

Definition (NSPACE): $L \in \text{NSPACE}(S(n))$ if there exists a verifier TM V and a constant $c \in \mathbb{N}$ such that for all x , $L(x) = 1$ if and only if there exists a witness $w \in \{0, 1\}^*$ such that $V(x, w) = 1$ using at most $c \cdot S(n)$ space, where $n = |x|$.

Note: The witness can be arbitrarily large as long as it is finite.

4.0.1 Can any problem be solved in NSPACE if the witness is not bounded? (e.g., could w just be the truth table for L ?)

Imagine we are trying to solve function $f : \{0, 1\}^{|x|} \rightarrow \{0, 1\}$ and we just set the witness w to the exact truth table for f because the size of w has no bound $2^{|x|}$ is still finite. The initial thought is we can use that to solve the function in $O(1)$ space, however we could construct a malicious or bad witness to trick this verifier that looks like a truth table to f , but is wrong on some or most values.

This verifier will work for the positive case where the witness is faithful, but it will fail on negative cases. The verifier must be able to reject a maliciously crafted "bad" w . The witness in NSPACE represents the correct sequence of non-deterministic branches, which has no direct time bound but is constrained by the graph of configurations.

Space is much more precious than time!

5 Space Hierarchy Theorem and PSPACE

Space Hierarchy Theorem: If $f(n) = o(g(n))$, where $f(n), g(n) \geq \log(n)$ and they are space-constructible, then:

$$\text{SPACE}(f(n)) \subsetneq \text{SPACE}(g(n))$$

The proof uses diagonalization and the construction of a Universal Turing Machine (UTM) that simulates machines while counting space.

PSPACE Classes

$$\text{PSPACE} = \bigcup_{c>0} \text{SPACE}(n^c) \subseteq \text{EXP}$$

$$\text{NPSPACE} = \bigcup_{c>0} \text{NSPACE}(n^c) \subseteq \text{EXP}$$

Example Problem in PSPACE:

$$\text{STMSAT}(\alpha, x, 1^n) = \begin{cases} 1 & \text{iff } M_\alpha(x) = 1 \text{ in space } n \\ 0 & \text{otherwise} \end{cases}$$

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References